

The one-dimensional frequently hypercyclic-subspace extension is false

Literature identification for arXiv:1311.7671

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Status: literature-implied full negative answer. A later theorem supplies a direct counterexample, although its authors do not identify the question in the source paper.

Source question

Muro, Pinasco and Savransky prove that, under their standing holomorphy-type hypotheses, every nontrivial convolution operator on $\mathcal{H}_{b\mathfrak{A}}(E)$ has a frequently hypercyclic subspace when $\dim E > 1$. Immediately before Theorem 4.3, in Section 4.1 on arXiv PDF page 13, they write:

The corresponding problem for $\dim(E) = 1$ is open, up to our knowledge.

Thus the question is whether the universal existence theorem remains true in one complex dimension [1].

Decisive later theorem

Let D be differentiation on the Fréchet space $H(\mathbb{C})$ of entire functions. Bayart, Ernst and Menet prove the following on supporting PDF page 3:

Theorem 1.2. Let P be a polynomial. Then $P(D)$ does not possess any frequently hypercyclic subspace.

See [2], Theorem 1.2 (and Theorem 3.2 for the full proof).

Identification and scope

Take $P(z) = z$. Then $P(D) = D$. Differentiation is a continuous nontrivial convolution operator on $H(\mathbb{C})$: it commutes with every translation and is not a scalar multiple of the identity. In one dimension, the source holomorphy-type construction gives the usual entire-function space $H(\mathbb{C})$. Hence D belongs to the source question's class but has no frequently hypercyclic subspace. The universal one-dimensional extension is therefore false.

This identification is an implication rather than an explicitly acknowledged answer. Bayart–Ernst–Menet state that they answer Bonilla and Grosse-Erdmann's earlier Problem 2; their paper does not cite Muro, Pinasco or Savransky. Accordingly this packet records a literature-implied answer, not a new counterexample and not an explicitly acknowledged literature answer.

Search evidence

The run’s registry, solution and attempt indexes were searched for the source arXiv id/title and the phrases “frequently hypercyclic subspace”, “convolution operator”, $H(\mathbb{C})$ and differentiation. A bounded web and arXiv search found the 2016 theorem above, the 2015 upper-frequent result, and later surveys confirming that $P(D)$ is frequently hypercyclic but admits no frequently hypercyclic subspace. The supporting PDF was checked directly.

References

- [1] S. Muro, D. Pinasco, and M. Savransky, *Strongly mixing convolution operators on Fréchet spaces of holomorphic functions*, Integral Equations Operator Theory 80 (2014), 453–468; arXiv:1311.7671.
- [2] F. Bayart, R. Ernst, and Q. Menet, *Non-existence of frequently hypercyclic subspaces for $P(D)$* , Israel J. Math. 214 (2016), 149–166, doi:10.1007/s11856-016-1352-3.